Chapter 3: Expectation of Random Variables

AMS 310
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(2) Expectation and Variance of Continuous R.V.s

- Expected value (mean) of a continuous r.v.:

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

- Variance of a continuous r.v.:

$$\sigma^2 = Var(X) = E[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x) dx$$

- Standard Deviation of X: $\sigma = \sqrt{Var(X)}$

- $Var(X) = E(X^2) - [E(X)]^2$

Example 3.10:

The distribution of the amount of gravel (in tons) sold by a particular construction supply company in a given week is continuous with pdf

$$f(x) = \begin{cases} 6x(1-x), 0 \le x \le 1 \\ 0, \qquad \text{otherwise} \end{cases}$$

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^1 6x^2(1-x)dx \ = 6\int_0^1 (x^2 - x^3)dx$$

$$= 6\left(\frac{x^3}{3} - \frac{x^4}{4}\right)\Bigg|_0^1 = 6\left(\frac{1}{3} - \frac{1}{4}\right) = \frac{1}{2}$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^1 6x^3(1-x)dx$$

$$= 6\int_0^1 (x^3 - x^4)dx = 6\left(\frac{x^4}{4} - \frac{x^5}{5}\right)\Bigg|_0^1 = 6\left(\frac{1}{4} - \frac{1}{5}\right) = \frac{3}{10}$$

$$Var(X) = E(X^2) - [E(X)]^2 = \frac{3}{10} - \left(\frac{1}{2}\right)^2 = \frac{1}{20}$$

$X \sim \text{Uniform}(a, b)$

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_a^b \frac{x}{b-a} dx = \frac{x^2}{2(b-a)}\Bigg|_a^b = \frac{b^2 - a^2}{2(b-a)}$$
$$= \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2}$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_a^b \frac{x^2}{b-a} dx = \frac{x^3}{3(b-a)}\Bigg|_a^b = \frac{b^3 - a^3}{3(b-a)}$$
$$= \frac{(b-a)(b^2 + ab + a^2)}{3(b-a)} = \frac{a^2 + ab + b^2}{3}$$

$$Var(X) = E(X^2) - [E(X)]^2 = \frac{a^2 + ab + b^2}{3} - \left(\frac{a+b}{2}\right)^2$$
$$= \frac{4(a^2 + ab + b^2) - 3(a+b)^2}{12} = \frac{a^2 - 2ab + b^2}{12} = \frac{(a-b)^2}{12}$$

$X \sim$Unif(a, b), $a<b$

$$E(X) = \frac{a+b}{2}$$

$$\mathrm{Var}(X) = \frac{(b-a)^2}{12}$$

If $X \sim$Uniform$(0, 1)$,

$$E(X) = \frac{1}{2}$$

$$\mathrm{Var}(X) = \frac{1}{12}$$

Some Special Expectations

- n-th moment of X: $E(X^n)$
- n-th central moment of X: $E[(X-\mu)^n]$
- Variance (2nd central moment):
  $\sigma^2 = \mathrm{Var}(X) = E[(X-\mu)^2] = E(X^2) - \mu^2$
- If a and b are constants,
  $$E(aX+b) = aE(X) + b$$
- If a and b are constants,
  $$\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$$

Example 3.11

$$f(x) = \begin{cases} 3x^2, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

(a)

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^1 3x^3 dx = \left.\frac{3x^4}{4}\right|_0^1 = \frac{3}{4}$$

(b)

$$E(4X + 5X^2) = \int_{-\infty}^{\infty} (4x + 5x^2) f(x) dx$$

$$= \int_0^1 (12x^3 + 15x^4) dx = [3x^4 + 3x^5]_0^1$$

$$= 3 + 3 = 6$$

Theorem 3.1 For a nonnegative continuous r.v. X,

$$E(X) = \int_0^\infty P(X > x)dx$$

<u>Proof</u>

$$\int_0^\infty P(X > x)dx = \int_0^\infty [1 - F_X(x)]dx$$

$$= \int_0^\infty \int_x^\infty f_X(y)dy\,dx = \int_0^\infty \int_0^y f_X(y)dx\,dy$$

$$= \int_0^\infty \left(\int_0^y dx\right) f_X(y)dy = \int_0^\infty y f_X(y)dy = E(X)$$

Example 3.12

Let $X \sim$Unif(0, 1), and $Y = e^X$.

(a) Find the cdf $F_Y(y)$ of Y.

(b) Find the pdf $f_Y(y)$ of Y.

(c) Find $E(e^X)$.

Answer to Example 3.13

$X \sim$Unif(0, 1)

(a) $Y = e^X$

$$F_Y(y) = P(Y \le y) = P(e^X \le y) = P(X \le \ln y)$$

$$= \begin{cases} 0, & y \le 1 \\ \ln y\,, & 1 < y < e \\ 1, & \mathrm{y} \ge e \end{cases}$$

(b)

$$f_Y(y) = F_Y'(y) = \frac{1}{y}, \qquad 1 < y < e$$

(c)

$$E(e^X) = \int_{-\infty}^{\infty} e^x f(x) dx = \int_0^1 e^x \; dx = e^x \Big|_0^1 = e - 1$$

$$X \sim \mathrm{N}(\mu, \sigma^2)$$

$$E(X) = \mu, \qquad \mathrm{Var}(X) = \sigma^2$$

<u>Proof</u>

Let $Z = \frac{X-\mu}{\sigma}$. Then

$$E(X) = \int_{-\infty}^{\infty} \frac{x}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\, dx = \int_{-\infty}^{\infty} \frac{(\sigma z + \mu)}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\, dz$$

$$= \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z e^{-\frac{z^2}{2}}\, dz + \mu \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\, dz = 0 + \mu = \mu$$

$$\mathrm{Var}(X) = \int_{-\infty}^{\infty} \frac{(x-\mu)^2}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\, dx = \int_{-\infty}^{\infty} \frac{\sigma^2 z^2}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\, dz$$

$$= \sigma^2 \left[-\frac{z}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\, dz \right] = \sigma^2(0+1) = \sigma^2$$

- The cdf of $X \sim \text{Exponential}(\lambda)$ is

$$F(x) = \int_{-\infty}^{x} f(y)dy = \begin{cases} 0, & x \leq 0 \\ 1 - e^{-\lambda x}, & x > 0 \end{cases}$$

- $E(X) = \frac{1}{\lambda}$
- $\text{Var}(X) = \frac{1}{\lambda^2}$
- Survival function

$$S(x) = P(X > x) = 1 - F(x)$$

$$= \begin{cases} 1, & x \leq 0 \\ e^{-\lambda x}, & x > 0 \end{cases}$$

$X \sim$ Exponential(λ)

- $E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_{0}^{\infty} x\lambda e^{-\lambda x} dx$

$$= -xe^{-\lambda x}\Big|_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx = -\frac{1}{\lambda} e^{-\lambda x}\Big|_0^{\infty} = \frac{1}{\lambda}$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 \lambda e^{-\lambda x} dx$$

$$= -x^2 e^{-\lambda x}\Big|_0^{\infty} + \frac{2}{\lambda}\int_0^{\infty} x\lambda e^{-\lambda x} dx$$

$$= \frac{2}{\lambda} E(X) = \frac{2}{\lambda^2}$$

- $\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$

$X \sim \text{Gamma}(\alpha, \lambda),\ \alpha > 0, \lambda > 0$

$$E(X) = \alpha/\lambda,\ \text{Var}(X) = \alpha/\lambda^2 \text{ (check)}$$

- If $\alpha = 1$, $X \sim$ Exponential(λ)
- If $\alpha = n$, a positive integer, then X is the time until n events have occurred, when events occur according to a Poisson process with rate λ.

χ^2 Distribution with ν Degrees of Freedom (df)

- $X \sim \text{Gamma}(\nu/2, 1/2) = \chi^2(\nu)$

The pdf is

$$f(x) = \begin{cases} \dfrac{e^{-x/2} x^{(\nu/2)-1}}{\Gamma(\nu/2) 2^{\nu/2}}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = \nu, \qquad \text{Var}(X) = 2\nu$$

- Used for inferences about sample variance
- $Z \sim N(0,1) \Rightarrow X = Z^2 \sim \chi^2(1)$

Beta Distribution

- $X \sim \text{Beta}(a, b)$, $a > 0, b > 0$ with pdf

$$f(x) = \frac{1}{B(a,b)} x^{a-1}(1-x)^{b-1}, \qquad 0 < x < 1$$

$$B(a,b) = \int_0^1 x^{a-1}(1-x)^{b-1}\, dx = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

$$E(X) = \frac{a}{a+b}, \qquad \text{Var}(X) = \frac{ab}{(a+b)^2(a+b+1)}$$

- $\text{Beta}(1,1)$ is $\text{Unif}(0, 1)$.

<u>Example 3.13</u>: At a gas station, the fraction of customers who purchase supreme gas in a week is a r.v. with $\mathrm{Beta}(1, 4)$.

If the gas station serves an average of 850 customers per week, how many customers are expected to purchase supreme gas next week?

$$E(X) = \frac{\alpha}{\alpha + \beta} = \frac{1}{1 + 4} = 0.2$$

The expected number of such customers is $850(0.2) = 170$.

T Distribution (Gosset) with ν df

- $U \sim \mathrm{N}(0,1)$, $V \sim \chi^2(\nu)$, U and V are independent

$\Rightarrow T_\nu = \frac{U}{\sqrt{V/\nu}} \sim t(\nu)$: t distribution with ν df with pdf

$$f(t) = \frac{\Gamma((\nu+1)/2)}{\Gamma(\nu/2)} \frac{1}{\sqrt{\nu\pi}} \frac{1}{[1+(t^2/\nu)]^{(\nu+1)/2}}, -\infty < x < \infty$$

T_ν has first $\nu - 1$ moments.

$$\mathrm{E}(T_\nu) = 0 \text{ if } \nu > 1, \quad \mathrm{Var}(T_\nu) = \frac{\nu}{\nu - 2} \text{ if } \nu > 2$$

- $f(x)$: Bell-shaped and symmetric about 0
- The distribution has thicker tails than $N(0, 1)$.
- $\lim_{\nu \to \infty} T_\nu \to \mathrm{N}(0,1)$ in distribution
- Used for inferences about sample mean

Cauchy Distribution

- $X \sim$ Cauchy(θ) with pdf

$$f(x) = \frac{1}{\pi[1 + (x - \theta)^2]}, -\infty < x < \infty$$

- $\theta = 0 \Rightarrow X \sim t(1)$ with pdf

$$f(x) = \frac{1}{\pi(1 + x^2)}, -\infty < x < \infty$$

- Moment does not exist.

$$E(|X|) = \int_{-\infty}^{\infty} \frac{|x|}{\pi(1+x^2)} dx = \frac{2}{\pi} \int_0^{\infty} \frac{x}{1+x^2} dx$$

$$= \frac{1}{\pi} \ln(1 + x^2)\Big|_0^{\infty} = \infty$$